



# Normal Microbiota of GI tract

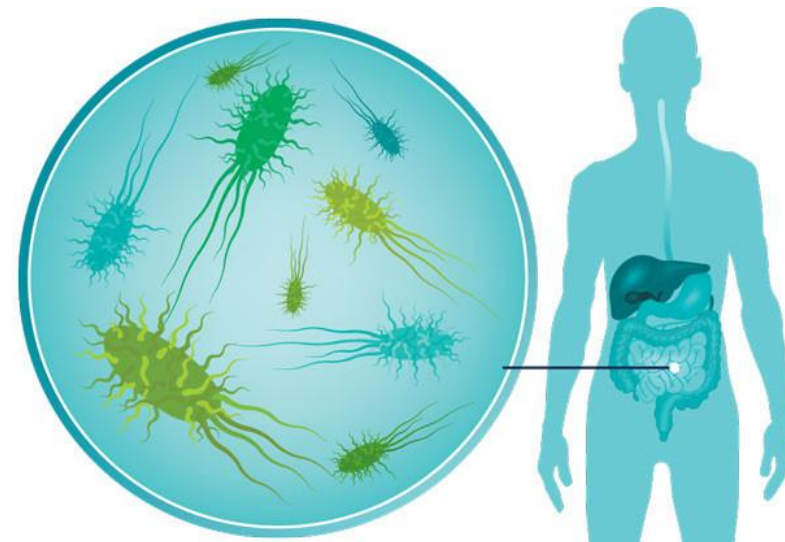
SGL 1

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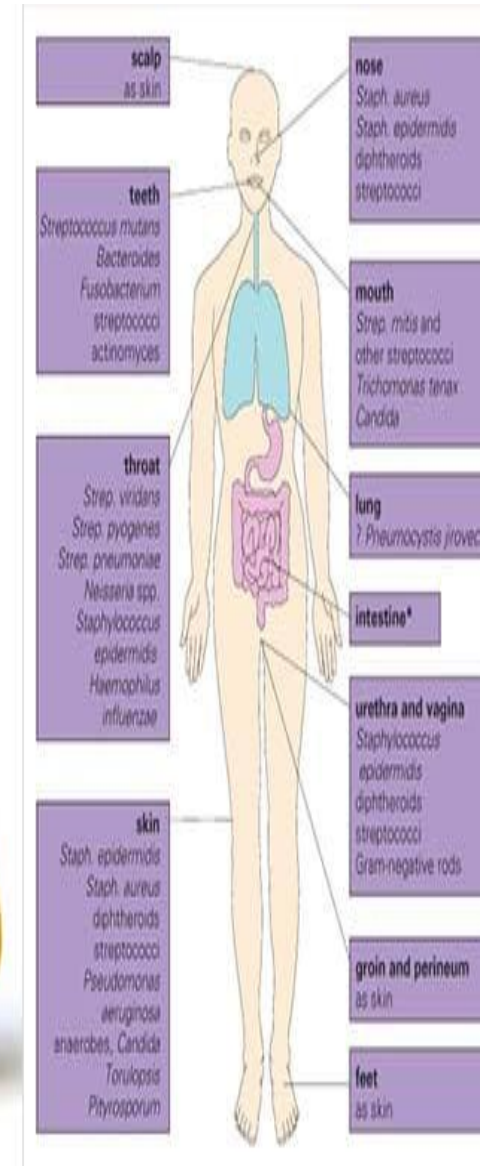
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# Normal Human Microbiota

- Population of **microorganisms** that inhabit the **skin** and **mucous membranes** of healthy normal persons.
- Their flourishing depends on physiologic factors of **temperature, moistures, and certain nutrients**

## Species of Microbiota

- Bacteria
- Fungi
- viruses



# When Does It Start?

- ❑ previously, it was believed that a healthy **newborn** enters the world in an essentially **sterile** condition. However, a growing number of scientists now believe that **fetuses are seeded with normal microbiota in utero**.
- ❑ These microbes are important **for healthy full-term pregnancies** and the **development** of healthy immune systems in newborns.
- ❑ The **genomes** of these microbial symbionts are collectively defined as the **microbiome**.



## Role of the resident microbiota

The skin and mucous membranes always harbour a variety of microorganisms that can be arranged into two groups:

- (1) **Resident microbiota** consists of relatively fixed types of microorganisms regularly found in each area at a given age; if it is disturbed, it promptly reestablishes itself.
- (2) **Transient microbiota** consists of non-pathogenic or potentially pathogenic microorganisms that inhabit the skin or mucous membranes for hours, days, or weeks.

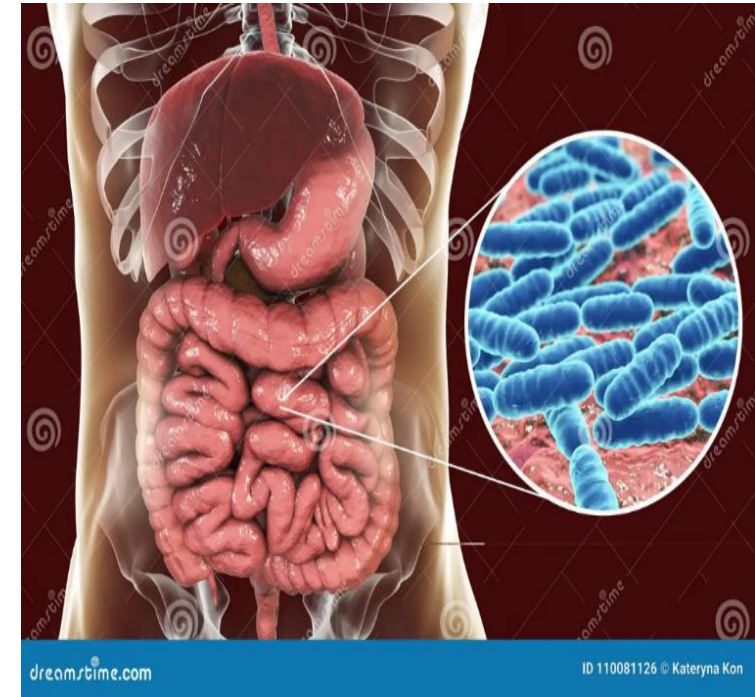
## Transient microbiota

- Derived from the **environment**
- Does not produce **disease**
- Does not establish itself **permanently** on the surface.
- Are generally of **little significance** so long as the normal resident flora remains intact.
- However, if the resident microbiota is disturbed, transient microorganisms may **colonize, proliferate, and produce disease.**

# Microbiota of the Gastrointestinal Tract

- ❑ The human gastrointestinal tract is **divided into sections**, allowing digestion and nutrient absorption in the proximal region to be separate from the **vast microbial populations** in the large intestine.
- ❑ **At birth**, the intestine is sterile, but organisms are soon introduced by food. The environment (**eg, maternal vaginal, fecal, or skin microbiota**) is a major factor in determining the **early microbial profile**.
- The intestinal microbiota of **breastfed children** is dominated by **Bifidobacteria**.
- In **bottle-fed children**, a more mixed flora exists in the bowel, and **lactobacilli** are less prominent.

- ❑ **As food habits develop** toward the adult pattern, the bowel flora changes. **Diet** has a marked influence on the relative composition of the intestinal and fecal flora. For example,
- Individuals on an animal-based diet have been shown to have
    - **Increased abundance** of bile-tolerant microorganisms (*Alistipes*, *Bilophila*, and ***Bacteroides***)
    - **Decreased levels** of **Firmicutes** that metabolize dietary plant polysaccharides (*Roseburia*, *Eubacterium rectale*, and *Ruminococcus bromii*).



- In normal adults, the **esophagus** contains microorganisms arriving with saliva and food.
- **Stomach** receive large number of microorganism from mouth along with food and water. In an adult, the density of microorganisms in the stomach is relatively low ( $10^2$  to  $10^3$  per gram of contents) due to gastric enzymes and acidic pH.
- From the hundreds of phylotypes detected in the human stomach, only *Helicobacter pylori* persists in this environment.



- The normal acid pH of the stomach markedly protects against infection with some enteric pathogens (eg, *Vibrio cholerae*).
- Administration of **antacids**, **H<sub>2</sub>-receptor antagonists**, and gastroprotection inhibitors for peptic ulcer disease and gastroesophageal reflux disease leads to a great **increase in microbial flora of the stomach**, including many organisms usually prevalent in feces.
- As the pH of intestinal contents becomes alkaline, the **resident flora gradually increases**.

- **In the adult duodenum**, there are  $10^3$ – $10^4$  bacteria/mL of effluent,
- with higher populations in the **jejunum**,  $10^4$ – $10^5$  bacteria/mL,
- **Ileum**,  $10^6$ – $10^7$  bacteria/mL;
- in the **cecum and transverse colon**,  $10^{11}$ – $10^{12}$  bacteria/mL, which is the highest record for any microbial habitat.
- In the upper intestine, the bacterial population associated with the mucosa includes the phylum; **Bacteroidetes and members of the Clostridiales**.
- Lumen and cecum; **members of the Enterobacterales and enterococci**.
- Sigmoid colon and rectum, the bacteria constitute about **60% of the fecal mass**.
- **Anaerobes** outnumber facultative organisms by 1000-fold.
- In **diarrhea**, the bacterial content may diminish greatly, but in **intestinal stasis**, the count rises.

❑ In a normal adult colon, 96–99% of the resident bacterial flora consists of **anaerobes**. **Six major phyla predominate;**

- Bacteroidetes
- Firmicutes
- Actinobacteria
- Verrucomicrobiota,
- Fusobacteria
- Proteobacteria.

❑ More than **1000 distinct types of organisms**, which can be cultured routinely in the laboratory, occur regularly in normal fecal flora.

- **Archaea** are represented primarily by the methane producer ***Methanobrevibacter smithii***, a low-abundance microorganism that may play an important role in stabilizing gut microbial communities.
  - More than 500 species of **bacteria** in the colon.
  - **protozoans and fungi**, whose functions are less well understood.
  - **Viruses**, mostly phages whose hosts are prominent members of the microbiota, are remarkably common in the colon.
- ❑ Minor trauma (eg, sigmoidoscopy, barium enema) may induce transient bacteremia in about 10% of procedures.

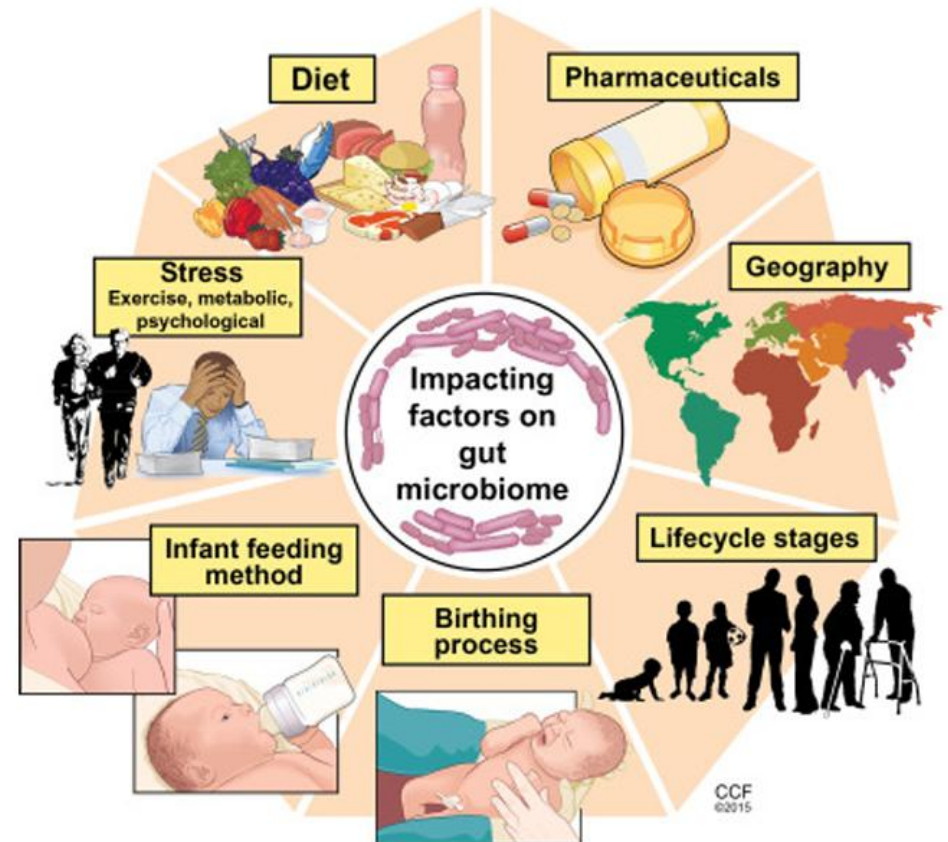
## The most common Bacteroidetes species and their roles in gut health

| Bacteria                                    | Effect                             | Mechanism of Action                                                                                                                   |
|---------------------------------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <i>Bacteroides fragilis</i> [79]            | Pathogenic                         | Production of <i>B. fragilis</i> toxin (BFT), leading to multidrug-resistant (MDR) infections in anatomical tissues                   |
| <i>Bacteroides thetaiotaomicron</i> [80,81] | Anti-inflammatory, B12 transporter | Suppression of <i>C. difficile</i> toxin production, promotion of gut barrier regeneration                                            |
| <i>Bacteroides vulgatus</i> [82]            | Inflammatory                       | Decrease in serum valeric acid (VA) and increase in osteoclast activity, activating pro-inflammatory pathways                         |
| <i>Bacteroides uniformis</i> [83,84]        | Antilipogenic                      | Degrading plant polysaccharides such as xyloglucan, suppressing ferroptosis in the liver and reducing free fatty acids in circulation |
| <i>Bacteroides ovatus</i> [85]              | Therapeutic, immune support        | Synthesizing the inhibitory neurotransmitter gamma-aminobutyric acid (GABA), producing SCFA, reducing tryptophan and glutamine levels |
| <i>Prevotella copri</i> [86]                | Immune support                     | Producing SCFA (except propionate) by fermenting complex carbohydrates in the intestine                                               |
| <i>Parabacteroides distasonis</i> [87]      | Anti-inflammatory                  | Reduction in inflammatory cytokines with S-layer containing glycoprotein, succinic acid production                                    |
| <i>Alistipes finegoldii</i> [76]            | Protective against colitis         | Synthesis of membrane lipids using exogenous fatty acids found in the intestine.                                                      |

## Factors that influences number and types of normal flora in GI tracts:

❑ **Diet:** Types of foods affects number and types of microflora in GI tract. In **vegetarian diet** the number of sacchrolytic microorganism such as *E. coli* increases and if **meat is consumed** proteolytic organism such as *Clostridium*

❑ **HCl and Bile:** HCl in stomach and bile in duodenum are antimicrobial so the number of microorganisms is lower in stomach and duodenum than other GI tract.



## Factors that influences number and types of normal flora in GI tracts:

- ❑ **Physiological condition:** during diarrhea number of microbiota decreases due to washing of intestinal tract.
- ❑ **The mode of delivery** (vaginal or caesarean)
- ❑ **Diet during infancy** (breast milk or formula feeds) and adulthood (vegan based or meat based)
- ❑ **Antibiotics:** if antibiotics is taken orally the number of microorganism decreases. Prolong antibiotic therapy decreases significant number of microflora of GI tract.

# Functions of intestinal microbiota

**The important functions of intestinal microbiota can be divided into three major categories:**

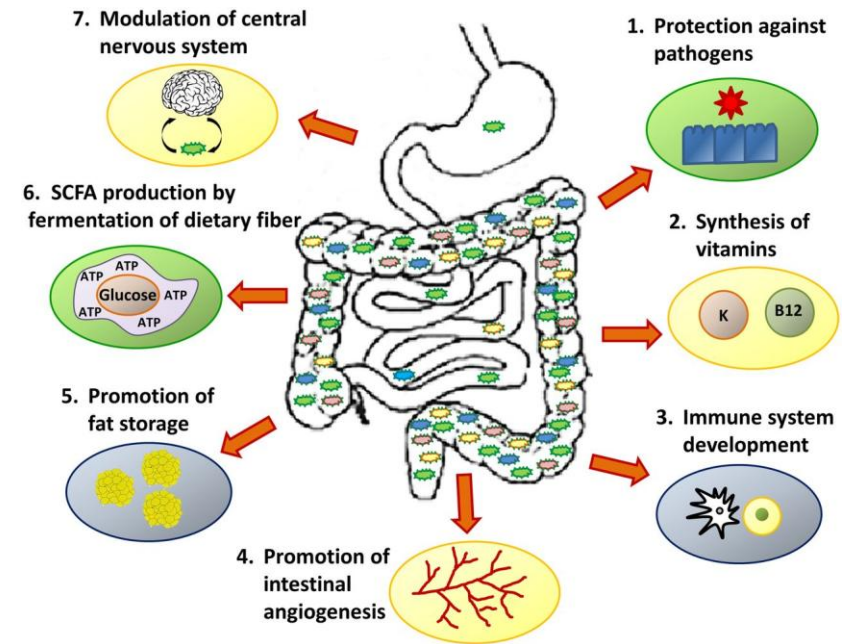
- ❑ The resident bacteria displace and inhibit potential pathogens
  - Competing for nutrients and receptors.
  - Production of **antimicrobial factors**, such as bacteriocins and lactic acid.
- ❑ Commensal organisms are important for
  - Development and function of the mucosal immune system.
  - They induce the secretion of IgA.

# Functions of intestinal microbiota

## □ A broad range of metabolic functions.

- The microbiota of the small intestine can contribute to the amino acid requirements of the host if they are not provided by the diet itself.
- They synthesize vitamin K, biotin, and folate and enhance ion absorption.
- Certain bacteria metabolize dietary carcinogens and assist with fermentation of nondigestible dietary residue.

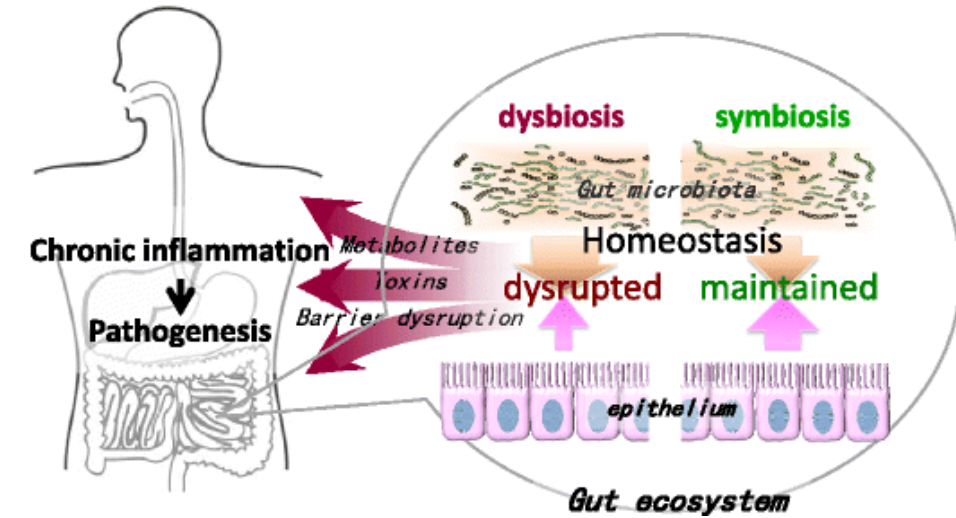
**Note.** The normal biota in the gut plays an important role in “teaching” our immune system to react to microbial antigens.



# Microbiota in health and diseases

Intestinal microbial balance is closely relevant to human diseases and health

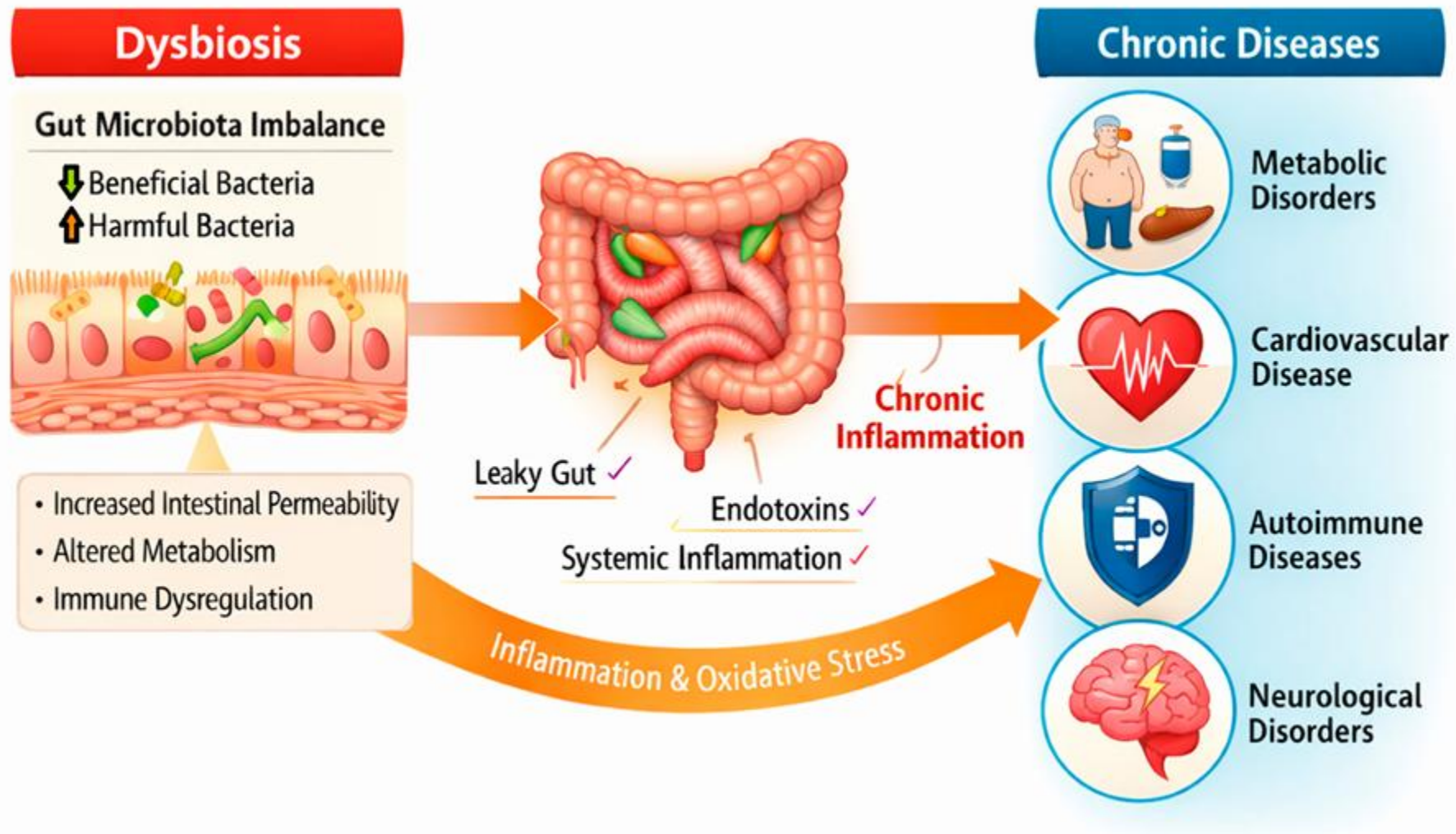
- ❑ The intestinal microbiota has an important influence on the metabolic processes. However, these beneficial conditions can be observed only in **symbiotic situations**, that is, when there is a **balance between commensal and pathogenic microorganisms**.
- ❑ **Symbiosis** describes the type of interaction between organisms. **Homeostasis** describes the stable condition that can result from that interaction.



# Symbiosis and Dysbiosis

- With **symbiosis**, the host harbors a healthy balance of each microbial community in a manner that is beneficial to the host. However, dysbiosis can also occur.
- **Dysbiosis**, also called dysbacteriosis, occurs when there is an imbalance (either in number or type) of microbial communities that have colonized the host.
- The fact that even **healthy people** harbor small numbers of **pathogenic** microbes is the growing problem of “*Clostridioides difficile* ” infections. *Clostridioides difficile* lives in the guts in very **small quantities**, kept in check by the microbial **antagonism** a diverse microbiota provides. When the healthy microbiota is **damaged**, especially through **broad-spectrum** or **long-term antibiotic** usage, *C. difficile* can **flourish**, leading to severe and long lasting intestinal disease.

- The anaerobic flora of **the colon**, including ***Bacteroides fragilis***, clostridia, and peptostreptococci, plays a main role in **abscess formation** originating in **perforation** of the bowel.
- In genetically susceptible individuals, some components of the flora can result in disease. For example, **inflammatory bowel diseases** are believed to be associated with a **loss of immune tolerance to bacterial antigens**. This leads to intense inflammation caused by an exuberant immune response. Similar mechanisms may be important in intestinal malignancy such as colon cancer.
- Exposure to aggressive factors such as **stress, excessive consumption of alcohol, smoking, antibiotic use and unhealthy diet** may cause imbalance of the microbiota, with increase of pathogenic bacteria and decrease of commensal bacteria.



Schematic illustration of the relationship between gut dysbiosis and chronic disease. A reduction in beneficial bacteria and an increase in harmful bacteria disrupt intestinal homeostasis, leading to increased intestinal permeability, endotoxin translocation, systemic inflammation, and oxidative stress, which may contribute to the development of metabolic, cardiovascular, autoimmune, and neurological disorders.

# Fecal microbiota transplantation (FMT)

- Fecal microbiota transplantation (FMT) also known as stool transplant is the process of transplantation of fecal bacteria **from a healthy individual into a recipient.**
- It has been used successfully as a treatment for patients suffering from ***Clostridium difficile infection*.**
- The hypothesis behind the success of FMT rests on the concept of bacterial interference, ie, **using harmless bacteria to displace pathogenic bacteria.**
- FMT restores the colonic microbiota to its natural state by replacing missing **Bacteroidetes and Firmicutes species.**



# Case Study

- A 70-year-old man with a history of diverticulosis of the sigmoid colon experiences a sudden onset of severe left lower quadrant abdominal pain. Fever develops. The severe pain gradually subsides and is replaced by a constant aching pain and marked abdominal tenderness. A diagnosis of probable ruptured diverticulum is made, and the patient is taken to the operating room. The diagnosis of ruptured diverticulum is confirmed, and an abscess next to the sigmoid colon is found. The most likely bacteria to be found in the abscess are
  - (A) Mixed normal gastrointestinal flora
  - (B) *Bacteroides fragilis* alone
  - (C) *Escherichia coli* alone
  - (D) *Clostridium perfringens* alone
  - (E) *Enterococcus* species alone

Q1/ The composition of the human microbiota remains relatively stable over time, unless there is a change in environmental conditions. Which term is used to describe this stability?

- A. Microbial symbiosis
- B. Microbial homeostasis
- C. Dysbiosis
- D. Colonisation resistance
- E. Microbial antagonism

Q2/ Antimicrobial therapy can decrease the amount of susceptible bowel flora and allow proliferation of relatively resistant colonic bacteria. Which one of the following species can proliferate and produce a toxin that causes diarrhea?

- A. *Enterococcus species*
- B. *Staphylococcus epidermidis*
- C. *Pseudomonas aeruginosa*
- D. *Clostridium difficile*
- E. *Bacteroides fragilis*

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